

## GEELONG, Australia

WMO: 948570

Lat: 38.1736S

Lon: 144.3764E

Elev: 13

StdP: 101.16

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 2.6        | 3.8 | 0.4                           | 3.9 | 9.9  | 1.6 | 4.2 | 8.9  | 12.4                  | 12.3 | 11.4 | 11.6 | 1.7                   | 290  | 0.553 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 9.9                    | 34.0            | 18.9 | 30.8 | 18.6 | 27.7 | 17.9 | 21.2                | 27.6 | 20.2 | 26.2 | 19.4 | 25.0 | 6.2                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.5                            | 14.3 | 22.3 | 18.5 | 13.4 | 21.8 | 17.6 | 12.6 | 21.1 | 61.7          | 27.8 | 58.1 | 26.2 | 55.2 | 25.1 | 26.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (d)                        | (e)  | (f)                | (g) | (h)                                                | (i)  | (j)        | (k)  | (l)        | (m)  | (n)        | (o)  |    |    |
| 11.5              | 10.1 | 9.0 | -0.2                       | 40.2 | 1.2                | 2.5 | -1.1                                               | 42.0 | -1.8       | 43.4 | -2.5       | 44.8 | -3.4       | 46.6 |    |    |
|                   |      |     | -0.5                       | 23.2 | 1.2                | 1.2 | -1.4                                               | 24.1 | -2.1       | 24.8 | -2.7       | 25.5 | -3.6       | 26.4 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 14.1   | 18.8  | 19.1  | 17.6  | 14.8  | 12.4  | 10.3  | 9.8   | 10.3  | 11.7  | 13.3  | 15.1  | 16.9  |
|                                                                   | DBStd       | 4.34   | 3.71  | 3.45  | 3.37  | 2.82  | 2.23  | 1.96  | 1.88  | 2.06  | 2.56  | 3.17  | 3.40  | 3.63  |
|                                                                   | HDD10.0     | 81     | 0     | 0     | 0     | 1     | 3     | 18    | 26    | 21    | 8     | 3     | 1     | 0     |
|                                                                   | HDD18.3     | 1716   | 37    | 27    | 56    | 113   | 183   | 239   | 264   | 250   | 201   | 162   | 109   | 74    |
|                                                                   | CDD10.0     | 1591   | 272   | 254   | 235   | 143   | 79    | 29    | 21    | 29    | 58    | 104   | 154   | 212   |
|                                                                   | CDD18.3     | 186    | 50    | 48    | 32    | 7     | 0     | 0     | 0     | 0     | 1     | 6     | 13    | 28    |
|                                                                   | CDH23.3     | 1944   | 495   | 420   | 303   | 71    | 2     | 0     | 0     | 0     | 11    | 107   | 184   | 351   |
|                                                                   | CDH26.7     | 872    | 253   | 193   | 129   | 17    | 0     | 0     | 0     | 0     | 1     | 34    | 71    | 175   |
| Wind                                                              | WSAvg       | 4.1    | 4.5   | 4.3   | 4.0   | 3.6   | 3.7   | 3.7   | 4.1   | 4.3   | 4.3   | 4.3   | 4.3   | 4.4   |
| Precipitation                                                     | PrecAvg     | 575    | 39    | 36    | 37    | 46    | 51    | 46    | 49    | 54    | 53    | 60    | 48    | 45    |
|                                                                   | PrecMax     | 908    | 132   | 193   | 122   | 144   | 100   | 107   | 108   | 99    | 105   | 147   | 153   | 121   |
|                                                                   | PrecMin     | 282    | 1     | 1     | 5     | 8     | 8     | 9     | 10    | 2     | 15    | 6     | 5     | 2     |
|                                                                   | PrecStd     | 127    | 29    | 43    | 26    | 31    | 27    | 24    | 23    | 22    | 24    | 32    | 30    | 29    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 38.8  | 36.4  | 34.9  | 28.9  | 22.8  | 17.7  | 17.6  | 20.5  | 24.7  | 30.7  | 32.7  | 36.7  |
|                                                                   |             | MCWB   | 20.3  | 19.6  | 19.2  | 15.9  | 14.3  | 13.2  | 11.9  | 13.6  | 15.1  | 16.2  | 18.3  | 18.8  |
|                                                                   | 2%          | DB     | 33.3  | 32.5  | 30.2  | 25.2  | 20.1  | 16.0  | 15.6  | 17.7  | 21.2  | 26.2  | 28.5  | 31.7  |
|                                                                   |             | MCWB   | 19.6  | 19.1  | 18.6  | 15.7  | 14.1  | 12.4  | 11.3  | 11.9  | 13.9  | 15.2  | 17.8  | 18.3  |
|                                                                   | 5%          | DB     | 28.7  | 28.3  | 26.6  | 22.4  | 18.1  | 14.9  | 14.4  | 15.8  | 19.0  | 22.7  | 24.6  | 27.0  |
|                                                                   |             | MCWB   | 19.0  | 19.1  | 17.8  | 15.4  | 13.5  | 11.9  | 10.8  | 11.4  | 13.3  | 14.8  | 16.9  | 17.7  |
|                                                                   | 10%         | DB     | 24.8  | 24.9  | 23.2  | 20.0  | 16.6  | 13.9  | 13.3  | 14.4  | 16.9  | 19.6  | 21.4  | 23.2  |
|                                                                   |             | MCWB   | 18.6  | 18.7  | 17.5  | 15.0  | 13.1  | 11.4  | 10.4  | 10.7  | 12.3  | 13.9  | 16.1  | 17.2  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 22.9  | 22.3  | 21.0  | 18.1  | 16.4  | 14.5  | 13.2  | 14.4  | 16.5  | 18.1  | 20.8  | 21.3  |
|                                                                   |             | MCDB   | 28.5  | 28.5  | 27.7  | 23.8  | 18.5  | 16.0  | 15.5  | 18.5  | 21.1  | 24.4  | 26.9  | 28.2  |
|                                                                   | 2%          | WB     | 21.2  | 21.2  | 19.9  | 17.1  | 15.1  | 13.2  | 12.2  | 12.8  | 15.0  | 16.7  | 19.0  | 19.7  |
|                                                                   |             | MCDB   | 28.2  | 26.8  | 25.9  | 21.3  | 18.3  | 15.0  | 14.3  | 16.0  | 19.5  | 22.6  | 25.3  | 26.8  |
|                                                                   | 5%          | WB     | 20.0  | 20.1  | 19.0  | 16.3  | 14.2  | 12.4  | 11.4  | 11.9  | 13.8  | 15.5  | 17.6  | 18.5  |
|                                                                   |             | MCDB   | 26.5  | 25.0  | 24.0  | 20.5  | 17.1  | 14.3  | 13.5  | 14.8  | 17.8  | 21.0  | 23.0  | 24.8  |
|                                                                   | 10%         | WB     | 19.0  | 19.2  | 18.0  | 15.5  | 13.5  | 11.7  | 10.8  | 11.1  | 12.7  | 14.3  | 16.4  | 17.5  |
|                                                                   |             | MCDB   | 24.6  | 24.1  | 22.5  | 19.5  | 16.1  | 13.6  | 12.8  | 13.8  | 16.2  | 18.9  | 20.9  | 22.8  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.2  | 9.9   | 9.8   | 9.5   | 7.9   | 7.0   | 7.0   | 8.0   | 9.1   | 10.3  | 10.1  | 10.7  |
|                                                                   |             | MCDBR  | 18.5  | 17.2  | 16.3  | 14.2  | 10.5  | 8.4   | 8.1   | 10.5  | 12.9  | 17.1  | 17.1  | 19.0  |
|                                                                   |             | MCWBR  | 6.1   | 5.8   | 6.0   | 6.3   | 5.8   | 5.5   | 4.9   | 5.9   | 6.7   | 7.5   | 7.2   | 6.9   |
|                                                                   | 5% WB       | MCDBR  | 14.2  | 12.5  | 12.6  | 11.6  | 8.9   | 7.6   | 7.2   | 9.3   | 10.9  | 13.6  | 14.1  | 14.5  |
|                                                                   |             | MCWBR  | 5.9   | 5.5   | 5.5   | 5.9   | 5.6   | 5.4   | 4.9   | 5.8   | 6.4   | 7.0   | 6.9   | 6.3   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.352 | 0.355 | 0.335 | 0.327 | 0.314 | 0.301 | 0.303 | 0.312 | 0.333 | 0.339 | 0.347 | 0.343 |
|                                                                   | taud        |        | 2.509 | 2.510 | 2.565 | 2.564 | 2.574 | 2.602 | 2.587 | 2.558 | 2.489 | 2.488 | 2.493 | 2.525 |
|                                                                   | Ebn at Noon |        | 980   | 953   | 933   | 878   | 827   | 812   | 832   | 881   | 919   | 958   | 978   | 994   |
|                                                                   | Edh at Noon |        | 112   | 107   | 93    | 82    | 70    | 63    | 68    | 80    | 99    | 108   | 113   | 111   |
| All-Sky Solar Radiation                                           | RadAvg      |        | 7.08  | 6.19  | 4.87  | 3.40  | 2.27  | 1.79  | 2.00  | 2.81  | 4.03  | 5.37  | 6.25  | 7.04  |
|                                                                   | RadStd      |        | 0.43  | 0.28  | 0.28  | 0.18  | 0.11  | 0.13  | 0.11  | 0.16  | 0.25  | 0.39  | 0.36  | 0.41  |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) | Station Only            | +0.69 | +0.45   | +0.60  | N/A   | +0.53   | +0.65 | N/A | -198        | +243    | +53     |         |
| (47) | Regional (36 neighbors) | +0.49 | N/A     | N/A    | +0.84 | N/A     | N/A   | -15 | -129        | +173    | +54     |         |

Nomenclature: See separate page